

Recoverability of causal structure from psychological networks?

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1 Introduction

Psychological network models represent symptoms as nodes and their associations as edges, while the theories that motivate them concern something different: the processes by which one symptom brings about another. The gap between the two rarely matters until a cross-sectional network is read as a map of where to intervene, at which point an association is being asked to carry a causal claim it was never estimated to support. This project asks how much causal structure the cross-sectional datasets collected in ReBayesed can actually determine, and what, in the data or in the methods, sets the limit on orienting an edge.

Research problem. In these networks an edge is a partial correlation: an association that survives conditioning on every other measured symptom (Borsboom and Cramer, 2013). Causal discovery narrows the possible orientations of such edges using the pattern of conditional independencies, but observational data typically leave several structures compatible with the same pattern. How much ambiguity remains has not been assessed systematically across a published corpus of psychological networks.

Relevance. Undirected networks are routinely used to discuss intervention, yet intervention requires direction. The ReBayesed export shows how thin the evidence often is even for the undirected edges: in the tenth of networks with the least data per parameter, 56.7% of edges have inconclusive Bayesian evidence, against 17.1% in the tenth with the most. If an edge cannot be established as present, its direction is unlikely to fare better, and I will test whether that expectation holds.

Significance. Of the 294 networks, 147 have fewer than 5 observations per edge parameter and 44 have fewer than 1. If orientation tracks data volume, the result gives sample-planning guidance for future studies; if orientation stays ambiguous even where edge evidence is strong, the result identifies a limit that more data would not remove. Either outcome is informative.

Novelty. Huth et al. (2026) quantified the evidence for undirected edges; Huber (2024) set out why causal structure is often not identifiable from observational data. Neither applied the other’s question to this corpus. This project does.

2 Related work

Huth et al. (2026) re-analysed 294 cross-sectional psychological networks from 126 papers with Bayesian model averaging and showed that edge evidence depends strongly on sample size. Their export is the input here, because it lets me ask of the same datasets whether the evidence that supports an edge also supports a direction.

Huber (2024) explains why observational data can leave causal structures indistinguishable, particularly when common causes are unmeasured. I use this framework to fix the assumptions under which any orientation the analysis produces should be read.

For the analysis I will use `causal-learn` (Zheng et al., 2024), which implements both PC and FCI in Python. PC assumes that all common causes are measured; FCI drops that assumption at the cost of weaker conclusions. Running both from one library keeps the comparison in RQ3 consistent, since any difference then reflects the assumption rather than the implementation.

3 Research questions

Main question: how much causal structure can be recovered from these psychological datasets, and what limits that recovery?

1. Do networks with more observations per edge parameter yield a greater proportion of determined edge orientations?
2. Does stronger Bayesian evidence for an edge translate into a determined direction for that edge?
3. How do the conclusions change when unmeasured common causes are allowed?

As a supplementary comparison, I will check whether independent studies that measured the same variables agree on orientation.

4 Expected outcomes

Approach. The export contains correlation matrices, sample sizes and metadata rather than participant-level observations, so the first task is to confirm that Fisher’s z tests, and `causal-learn` on top of them, can run from a correlation matrix and a sample size alone. The export also needs cleaning before anything runs on it: 233 of the 294 networks have variable names that disagree with their matrix labels, and the export holds 294 networks where the article reports 293. The data are 71 continuous, 196 ordinal and 27 mixed networks. Exploratory analysis of the export is complete and is the source of these counts; the cleaning itself has not started. A pilot in week 5 will decide which continuous networks form the analysis subset and which test settings to fix.

RQ1: data volume and orientation. I will run PC on each network in the subset, record its adjacencies and the proportion of edges it orients, and then check how sensitive those proportions are to the significance threshold and to variable ordering. Because 58 of the 126 papers report several networks that are potentially dependent, I will select one network per paper by a prespecified rule and report the others separately, so that dependent networks do not inflate the trend. I expect the oriented proportion to rise with observations per edge parameter but to stay well short of complete, which is the pattern that would show data volume matters without being the whole story.

RQ2: statistical evidence and direction. I will relate each network’s oriented proportion to its proportion of inconclusive Bayesian edges under each of the three priors. Bayesian edge evidence enters as a covariate rather than as ground truth, because a partial-correlation edge and a causal adjacency are not the same object: an edge can be estimated with confidence and still leave its direction undetermined. Those edges, strong evidence with no direction, are the ones I will examine most closely, since they mark where the limit is set by identifiability rather than by sample size.

RQ3: allowing unmeasured common causes. I will run FCI on the same subset. FCI and PC do not output the same objects, so counting arrowheads would compare unlike things; instead I will compare statements both can make, such as whether one variable is an ancestor of another, and quantify how much of PC’s structure dissolves once unmeasured common causes are permitted. Both algorithms still assume acyclicity, the Markov and faithfulness conditions, and the absence of selection bias (Huber, 2024), and these assumptions bound what any result in this project can claim.

Cross-study comparison and outputs. Depression is the largest subtopic, with 32 networks, and the first candidate for comparing orientations across studies; whether that comparison is feasible depends on the metadata cleaning in week 4. Where studies disagree, I will distinguish a direction left unresolved from one that conflicts across studies, since only the second bears on replication. The deliverables are a cleaned pipeline, reproducible analyses, figures and a written interpretation that claims no more than the data determine under the assumptions above.

5 Research plan

The project runs for 13 weeks and is now in week 4. The first three weeks went to exploratory analysis of the export, which produced the descriptive figures above and exposed the cleaning problems; the plan below covers the remaining ten weeks. Data coordination is shared with the other students on the project; I lead the Python analyses and the interpretation. Literature review and writing run alongside the analysis rather than after it.

Week	Activity and output
4	Clean metadata; resolve the 294–293 discrepancy; identify eligible networks and studies.
5	Pilot: validate Fisher’s z and <code>causal-learn</code> on correlation-matrix input; fix the subset and test settings.
6	Build and validate the PC pipeline; run PC on the subset.
7	Summarise adjacencies and orientations; analyse data volume and sensitivity (RQ1).
8	Relate orientation to Bayesian evidence (RQ2).
9	Run FCI on the matched subset.
10	Compare analyses with and without unmeasured causes (RQ3).
11	Complete the cross-study comparison and checks; draft results.
12	Complete the thesis draft and discuss feedback.
13	Revise, check reproducibility and prepare the presentation.

The compression falls on weeks 6 and 11, where building the PC pipeline shares a week with running it and the cross-study comparison shares a week with the final checks. Both are workable if the pilot in week 5 settles the subset and settings, which is why that week is protected. Two further constraints are addressed early. If FCI proves computationally prohibitive, I will run

both algorithms on a smaller prespecified subset so that RQ3 survives intact. If no independent studies share variables, I will report that as a limitation of the corpus rather than substitute within-paper comparisons, which cannot count as replication. The subset is fixed by week 5.

References

- Denny Borsboom and Angélique O. J. Cramer. Network analysis: An integrative approach to the structure of psychopathology. *Annual Review of Clinical Psychology*, 9:91–121, 2013. doi: 10.1146/annurev-clinpsy-050212-185608.
- Martin Huber. An introduction to causal discovery. *Swiss Journal of Economics and Statistics*, 160(1):14, 2024. doi: 10.1186/s41937-024-00131-4.
- Karoline B. S. Huth, Jonas M. B. Haslbeck, Sara Keetelaar, Ruth J. van Holst, and Maarten Marsman. Statistical evidence in psychological networks. *Nature Human Behaviour*, 10(2): 333–346, 2026. doi: 10.1038/s41562-025-02314-2.
- Yujia Zheng, Biwei Huang, Wei Chen, Joseph Ramsey, Mingming Gong, Ruichu Cai, Shohei Shimizu, Peter Spirtes, and Kun Zhang. Causal-learn: Causal discovery in Python. *Journal of Machine Learning Research*, 25(60):1–8, 2024. URL <https://jmlr.org/papers/v25/23-0970.html>.